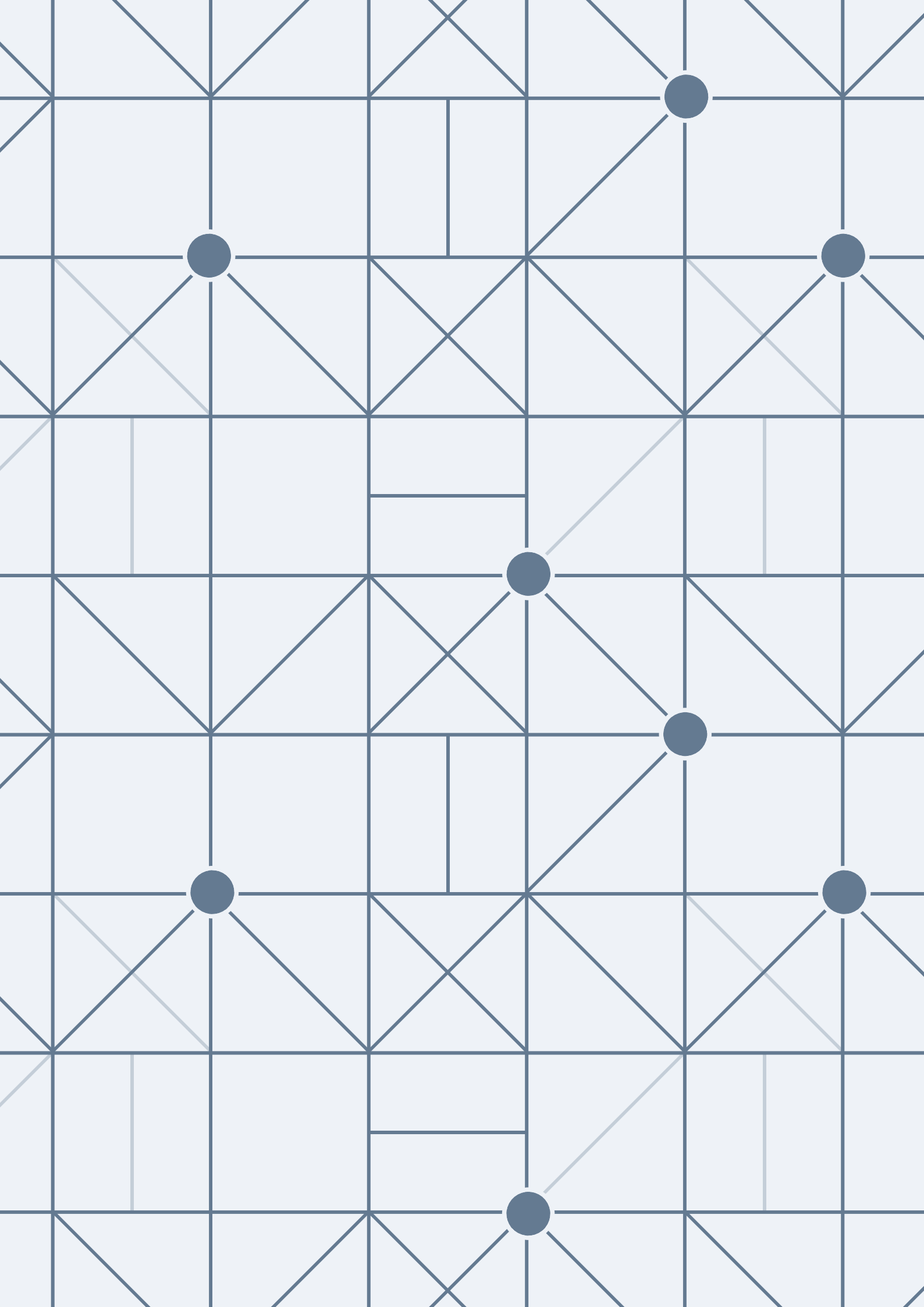




Microcanonical ensemble is finite $\leftarrow$ but this is not satisfied in any real system, it's always chaotic
then why do we study it? because it's the uncharted force to get the another ensembles.

POSTULATE OF AVERAGES

The **POSTULATE OF AVERAGES** states that, for a system in thermal equilibrium, the time average of a physical quantity over a sufficiently long period is equal to the ensemble average for the same quantity in the corresponding ensemble.

The **POSTULATE OF EQUAL A PRIORI PROBABILITY** asserts that, in the absence of any additional information or constraints, all microstates consistent with a given macrostate are equally probable.

The two postulates represent the **building blocks** with which we will construct the ensembles of thermodynamic systems in equilibrium, starting with the simplest case: the isolated system.

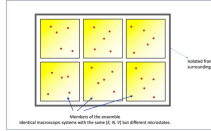
MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems identical copies of the original, **ISOLATED** and **IN EQUILIBRIUM**.

ISOLATED SYSTEM: a system that does not interact thermally, mechanically, or materially with the surroundings; characterised by *adiabatic, rigid, and impenetrable walls*

$V = cte$	$N = cte$	$H = E$
-----------	-----------	---------

IN EQUILIBRIUM: there is *no time dependence* of any variable, and the macrostate can be described using Thermodynamics. The density matrix/density function will also be independent of time.

Ensemble of isolated and equilibrium systems: **macrostate** (N, V, E)



MICROCANONICAL ENSEMBLE: the members of the ensemble are macroscopic systems identical copies of the original, **ISOLATED** and **IN EQUILIBRIUM**.

In the **classical** case, a microstate of the system is characterised by a point in phase space, Γ .

This phase space encompasses all possible combinations of positions and momenta for each particle in the system, and each point within this space represents a unique microstate.

$$(q, p) = (q_1, p_1, \dots, q_s, p_s) \longrightarrow 2s \text{ dimensions}$$

$\rho(\mathbf{q}, \mathbf{p}) \rightarrow$ probability density for the system to be in the microstate $(\mathbf{q}, \mathbf{p})$ (consistent with the macrostate)

What probability (statistical weight) do we assign to each of these microstates?

- **ISOLATED:** a system that does not interact thermally, mechanically, or materially with the surroundings.

$V = \text{const} \quad N = \text{const}$

$$H(\mathbf{q}, \mathbf{p}) = E = \text{const} \quad (q, p) \equiv (q_1, \dots, q_{3N}, p_1, \dots, p_{3N})$$

- **IN EQUILIBRIUM:** the density function is stationary.

$$\frac{\partial \rho}{\partial t} = 0 \quad \Rightarrow \quad \rho(q, p) = \rho(H(q, p))$$

WHAT IS THE DENSITY FUNCTION FOR THE MICROCANONICAL ENSEMBLE?



With the help of the postulate of equal a priori probability, it is possible to construct this density function clearly and directly.

The density function for the microcanonical ensemble is **constant** for all microstates consistent with the macrostate, assigning them equal probability.

In other words, for a system in a microcanonical equilibrium, **the density function is uniform in phase space**, reflecting the equal a priori probability of occupying any allowed microstate.

An isolated system with energy E is an idealisation.



- In reality, we can never completely eliminate interactions with the external world.
- Any small error in determining the energy covers a very broad spectrum of microstates.
- According to the Heisenberg Uncertainty Principle, obtaining the exact energy would require a measurement of infinite duration.

Instead of specifying the energy, we will define an isolated system as one whose energies fall within the interval ΔE .

$$\begin{aligned} [E, E + \Delta E] \\ \Delta E \ll E \end{aligned}$$

Diagram illustrating a thermodynamic cycle in the $H(q,p)$ vs q plane. The cycle is a closed loop with a blue outer boundary and a red inner boundary. The area between them is shaded grey. The top-left corner is labeled $H(q,p) = E + \Delta E$ and $\phi(E + \Delta E)$. The bottom-left corner is labeled $H(q,p) = E$ and $\phi(E)$. The right side is labeled $\Omega = -4\pi\epsilon$. Handwritten notes include: "increase in the temperature" with an arrow pointing to the top-left, "Amount of this hypothesis is E" pointing to the bottom-left, and "Amount of this hypothesis is E + ΔE" pointing to the top-left.

$$\Omega(E; \Delta E; N, V) = \Omega(E) = \phi(E + \Delta E) - \phi(E) \approx \frac{\partial \phi}{\partial E} \Delta E = \omega(E) \Delta E$$

We'll see that ΔF is

Le type approximation

$$\mathcal{Q}(C \subseteq \mathcal{Q}(E, N, V)) = \frac{\partial}{\partial C} \mathcal{Q} = u(C) \mathcal{Q} \rightarrow u(C) = \frac{\mathcal{Q}}{C} \rightarrow \text{number of molecules in } \mathcal{Q}$$

1. Any average over this ensemble must take into account that all mechanical states $(\mathbf{q}, \mathbf{p})$ for which $H \neq E$ cannot contribute to the mean value since they do not conserve energy.



2. ρ is a function **exclusively** of H .
3. **Equal a priori** probability for all microstates with $H = E$.
4. **Normalisation** condition: $\int \rho(\mathbf{q}, \mathbf{p}) d\mathbf{q}d\mathbf{p} = 1$

$$\rho(q, p) = \frac{1}{\omega(N, V, E)} \delta(E - H)$$

$$\omega(N, V, E) = \int \delta(E - H(q, p)) dq dp$$

Disc delta excludes
non-compatible

$$w(\text{NVC}) = \frac{\partial d}{\partial c} = \int_{c=0}^1 \sin(\pi x) dx = 1 \quad \text{mit Randwertfunktion } \rightarrow g=0 \quad \text{mit } x=0 \quad \text{mit } x=1$$

$$g(5,7) = g(11) = \frac{1}{2} d(c, 11)$$

$$p(x, \bar{x}) = \begin{cases} 1/2 & \text{exakte} \\ 0 & \text{sonst} \end{cases}$$

$$\omega(N, V, E) = \int \delta(E - H(q, p)) \, dq \, dp$$

We need a correction factor to adjust dimensions and avoid issues with entropy

A. This gives a volume, not a number of states!!
It needs correction $\rightarrow$ dividing by the "unit volume". $\rightarrow$

B. If particles are identical, swapping two particles does not give a new state

Divide by $N!$ → correct number of states

as sp^3 , the value that every single state has $\frac{\text{value}}{\text{value/state}} = n = \text{particle}$

whether this is sufficient, the entropy in the real world would be a measure - oriented. This means that larger N , the entropy do not increase

Final expressions

$$\rho(q, p) = \frac{1}{h^s N!} \frac{1}{\omega(N, V, E)} \delta(E - H)$$

$$\omega(N, V, E) = \frac{1}{h^s N!} \int \delta(E - H(q, p)) dq dp$$

Mean value of the dynamic function $\alpha(\mathbf{q}, \mathbf{p})$

$$\langle a(q, p) \rangle = \frac{1}{h^s N!} \frac{1}{\omega(N, V, E)} \int a(q, p) \delta(E - H(q, p)) dq dp$$

$$\langle a(q,p) \rangle = \int a(q,p) \rho(q,p) dq dp = \frac{1}{N} \frac{1}{\omega_C \omega_D} \int a(q,p) \delta(e - H(q,p)) dq dp$$

Ω is the most important quantity in the microcanonical ensemble.

From Ω , and its dependence on the variables N , V , and E , the **complete thermodynamics** of the system can be deduced.



What is the connection between Ω and thermodynamics?

Let's see that without $\frac{d}{ds} \int_{\Omega} f(s, p) ds dp = 0$.

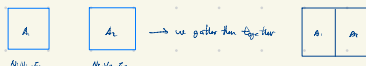
$$A = \int_V |\psi(\rho)|^2 d\Omega d\rho = \int_V \frac{1}{w} S(C-H) d\Omega d\rho = \frac{1}{w} \int_V f(C-H) d\Omega d\rho = \frac{\langle f|H\rangle}{\langle f|C-H|f \rangle} = \int_V f(C-H) d\Omega d\rho = h^2 N^2 \omega$$

$$\langle \sigma \rangle = \int_{\Omega} a(q, p) g(q, p) dq dp = \int_{\Omega} a(q, p) \frac{1}{h^N} \cdot \frac{1}{\omega} dC(H) d\zeta dp = \frac{1}{h^N \omega} \int_{\Omega} a(q, p) dC(H) d\zeta dp$$

Ins2 extensive function $\rightarrow$ $\text{Ins2} = \text{Ins2} \cdot \text{Ins2}$

1
this what is missing here

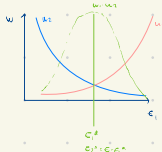
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$$X = G_0 \cup G_1 \cup G_2 \text{ where } G_i \cap G_j = \emptyset$$

The Hamiltonian A -H₀ whole system s.s.

$$\mathbf{H} = \mathbf{H}_1 + i\mathbf{H}_2$$

$$w = \frac{1}{N!N!} \int_0^1 \delta(C(t)) d\theta dp = \frac{1}{N!N!N_1!N_2!} \int_0^1 \delta(C(t_1, t_2)) d_1 dp_1 d_2 dp_2 = \frac{1}{N!} \int_0^1 d_1 dp_1 \frac{1}{N_1!} \int_0^1 \delta(C(t_1, t_2)) d_2 dp_2 \left. \begin{aligned} & \int_0^1 d_2 dp_2 \frac{1}{N_2!} \int_0^1 \delta(C(t_1, t_2)) d_1 dp_1 = \int_0^1 w(t_1) w(C(t_1)) dt_1 \\ & \int_0^1 \delta(C(t)) dt = \int_0^1 w(t_1) w(C(t_1)) dt_1 \end{aligned} \right\}$$



→ We know that $\int_0^1 u(x) \delta(x-c) dx = \int_0^1 u(x) \delta(x-c) e^{\frac{c-x}{2\sigma^2}} dx = u(c) \cdot u(c) e^{-\frac{c-c}{2\sigma^2}} = u(c) \cdot u(c) e^0 = u(c) \cdot u(c) = u(c)^2$ (Gaussian distribution integral)

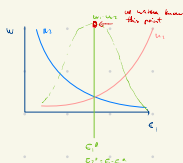
1. I estimated the height, the most contribution to the signal on the points near to the point's contribution are $\rightarrow \int_{-\infty}^{\infty} \delta$

$\mu(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

Let's maximize the likelihood:

$-\log L(\theta) = -\log \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_i^2} \right) = \frac{n}{2} \log(2\pi) + \frac{1}{2} \sum_{i=1}^n x_i^2$

This is a convex function, so the minimum is at the origin: $\theta = 0$.



We seek the equilibrium $\in$ that minimizes

[illegible]